6 Appendix

First we include extra theoretical details and proofs (section 6.1). Then we present further experimental details and examples (section 6.2).

6.1 Theory

First we briefly describe notation (section 6.1.1), present a formal definition of ignorance (section 6.1.2), and give the graphical criteria results for intention (section 6.1.3). Then we present the proofs of all technical results in the paper (section 6.1.4). In section 6.1.5, we state the PSO algorithm. Next we discuss intention, illustrating how our definition fixes problems with H&KW's definition. Then we briefly discuss the advantages and limitations of our definition of belief.

6.1.1 Notation

We use capital letters for variables (e.g., Y), lower case for their outcomes (e.g., y), and bold for sets of variables (e.g., $\boldsymbol{Y}$) and of outcomes (e.g., $\boldsymbol{y}$). We use $\text{dom}(Y)$ to denote the set of possible outcomes of variable Y, which is assumed finite. We use $\boldsymbol{Y} = \boldsymbol{y}$, for $\boldsymbol{Y} = \{Y_1, \ldots, Y_n\}$ and $\boldsymbol{y} = \{y_1, \ldots, y_n\}$, to indicate $Y_i = y_i$ for all $i \in \{1, \ldots, n\}$. For a set of variables $\boldsymbol{Y}$, $dom(\boldsymbol{Y}) = \bigtimes_{Y \in \boldsymbol{Y}} dom(Y)$ (i.e. the Cartesian product over domains). We use standard terminology for graphs and denote the parents of a variable Y with $\mathbf{Pa}_Y$.

6.1.2 Defining Ignorance

A well-known difficulty for Bayesian formulations of belief is distinguishing between a *false belief* and *ignorance* [61, 53, 9]. Our operalization of belief as acceptance provides a natural distinction between these cases. Following El Kassar [27], we define *ignorance* as a lack of true belief. When defining ignorance, as with belief, we require response so that the agent's ignorance can be inferred from their behaviour.

Definition 6.1 (Ignorance)**.** Under $\boldsymbol{\pi}$, in $\boldsymbol{e}$, for agent i and proposition ϕ to which i responds, i is *ignorant* about ϕ if they do not have a true belief about ϕ.

Example 6. Suppose a mugger T wishes to know S's secret pin number X. S can reveal the pin or stay silent, and T can guess the pin or give up. If S stays silent and T gives up, then S has caused T to be ignorant but has not caused a false belief.

6.1.3 Graphical criteria for intention

Theorem 6.2 (Soundness: intention)**.** *For any $\boldsymbol{\pi} = (\pi^i, \boldsymbol{\pi}^{-i})$, if i intentionally causes $X(\boldsymbol{\pi}, \boldsymbol{e})$, then there is a directed path from D^i to U^i passing through X in $\mathcal{G}$ (for some $U^i \in \boldsymbol{U}^i$).*

Theorem 6.3 (Completeness: intention)**.** *For any graph $\mathcal{G}$ with a directed path from D^i to U^i through X (for some $U^i \in \boldsymbol{U}^i$), there exists some parameterisation $\boldsymbol{\theta}$ s.t. for the SCG $\mathcal{M} = (\mathcal{G}, \boldsymbol{\theta})$, for some policy profile $\boldsymbol{\pi} = (\pi^i, \boldsymbol{\pi}^{-i})$ and some setting $\boldsymbol{e}$, i intentionally causes $X(\boldsymbol{\pi}, \boldsymbol{e})$ with π^i.*

6.1.4 Proofs

In this section we minimally state the proofs of all technical results in the paper.

Belief

Proof of proposition 3.2. For 2) we assume that i only adapts its policy if observing ϕ changes its subjective probability distribution (i.e., it only adapts if the observation of ϕ provides information: $\pi^i = \pi^i(\phi)$ whenever $P^{\boldsymbol{\pi}}(\cdot \mid pa_{D^i}) = P^{\boldsymbol{\pi}}(\cdot \mid pa_{D^i}, \phi)$. If there exists an observation edge (X, D^i) for all $X \in \boldsymbol{P}$, where $\boldsymbol{P}$ is the set of variables constituting the formula ϕ, then $\pi^i = \pi^i(\phi)$.

Proof. 1)

Suppose that i believes ϕ under $\boldsymbol{\pi}$ in $\boldsymbol{e}$. Then 1. $D^i(\boldsymbol{\pi}, \boldsymbol{e}) = D^i_{\phi=\top}(\boldsymbol{\pi}_{i(\phi)}, \boldsymbol{e})$. Further, 2. D^i responds to ϕ under $\boldsymbol{\pi}_{i(\phi)}$ in $\boldsymbol{e}$, so $D^i_{\phi=\top}(\boldsymbol{\pi}_{i(\phi)}, \boldsymbol{e}) \neq D^i_{\phi=\bot}(\boldsymbol{\pi}_{i(\phi)}, \boldsymbol{e})$. Hence,

$$D^i(\boldsymbol{\pi}, \boldsymbol{e}) \neq D^i_{\phi=\bot}(\boldsymbol{\pi}_{i(\phi)}, \boldsymbol{e}) = D^i_{\neg\phi=\top}(\boldsymbol{\pi}_{D^i(\neg\phi)}, \boldsymbol{e}).$$

So the first condition for belief fails for the proposition $\neg\phi$. This follows from the uniqueness of $\pi^i_D(\phi)$ and a consistency requirement between $\pi_D(\phi)$ and $\pi_D(\neg\phi)$ which enforces the final equality.

2)

Suppose i believes ϕ under $\boldsymbol{\pi}$ in $\boldsymbol{e}$ and that there exists an observation edge (X, D^i) for all $X \in \boldsymbol{P}$, where $\boldsymbol{P}$ is the set of variables constituting the formula ϕ. Then $\pi^i = \pi^i(\phi)$ and hence a) $D^i(\boldsymbol{\pi}, \boldsymbol{e}) = D^i(\boldsymbol{\pi}_{i(\phi)}, \boldsymbol{e})$. In addition, because i believes ϕ by supposition, we have b) $D^i(\boldsymbol{\pi}, \boldsymbol{e}) = D^i_{\phi=\top}(\boldsymbol{\pi}_{i(\phi)}, \boldsymbol{e}) \neq D^i_{\phi=\bot}(\boldsymbol{\pi}_{i(\phi)}, \boldsymbol{e})$. So, a) says that i does in fact observe whether ϕ is true or false and b) says that they act as though ϕ is true and would have acted differently if ϕ were false. Hence, ϕ is true and i does not have a false belief. □

Intention

Proof of proposition 3.5.

Proof. Suppose 1) $X(\boldsymbol{\pi}_1, \boldsymbol{e}) = X(\boldsymbol{\pi}_2, \boldsymbol{e})$ for all π^i_1 and π^i_2 with any fixed $\boldsymbol{\pi}^{-i}$. Suppose i intentionally causes $X(\boldsymbol{\pi}, \boldsymbol{e})$ with π^i wrt $\hat{\pi}^i$ so that the inequality in def. 3.4 holds for subset minimal $\boldsymbol{w}_X$. But remove $\boldsymbol{e}$ from $\boldsymbol{w}_X$ and the inequality still holds by 1). So $\boldsymbol{w}_X$ is not minimal and we have a contradiction. □

Proof of theorem 3.6. This uses def. 6.6.

Proof. Suppose agent i intentionally causes $\boldsymbol{X}(\boldsymbol{\pi}, \boldsymbol{e})$ with π^i w.r.t. $\hat{\pi}^i$. Now we check the three conditions for actual causality (def. 6.6). 1. Clearly $D^i(\boldsymbol{\pi}, \boldsymbol{e})$ and $\boldsymbol{X}(\boldsymbol{\pi}, \boldsymbol{e})$ obtain in $\boldsymbol{e}$. 2. Take $\boldsymbol{Z} = \{\}$ and $d' = D^i(\hat{\boldsymbol{\pi}}, \boldsymbol{e})$. $\boldsymbol{X}(\hat{\boldsymbol{\pi}}, \boldsymbol{e}) \neq \boldsymbol{X}(\boldsymbol{\pi}, \boldsymbol{e})$ otherwise $\boldsymbol{e}$ would not be in a minimal $\boldsymbol{w_X}$ satisfying the inequality in def. 3.4. Hence, 2. holds. $\{D^i\}$ is clearly a subset minimal set satisfying 1. and 2. since the empty set does not satisfy 2. □

Graphical Criteria

Proof of soundness result for intention (theorem 6.2).

Proof. First, suppose a) there is no path from X to U^i in $\mathcal{G}$ for any $U^i \in \boldsymbol{U}^i$. Suppose there exists $\boldsymbol{\pi} = (\pi^i, \boldsymbol{\pi}^{-i})$, $\hat{\pi}^i$, and $\boldsymbol{e}$ s.t. i intentionally causes $X(\boldsymbol{\pi}, \boldsymbol{e})$ with π^i w.r.t. $\hat{\pi}^i$. Then there exists subset-minimal $\boldsymbol{Y} \supseteq X$ and subset-minimal $\boldsymbol{w}_Y \subseteq dom(\boldsymbol{E})$ for each $Y \in \boldsymbol{Y}$ s.t. $\boldsymbol{e} \in \boldsymbol{w_X} := \bigcup_{Z \in \boldsymbol{X}} \boldsymbol{w}_Z$ s.t. the inequality in def 3.4 holds. But then, by a) and Markov compatibility, for $\boldsymbol{Y}' = \boldsymbol{Y} \setminus \{X\}$: $\sum_{U \in \boldsymbol{U}^i} \mathbb{E}_{(\hat{\pi}^i, \boldsymbol{\pi}^{-i})}[U_{\{Y_{\boldsymbol{\pi}|\boldsymbol{w}_Y}\}_{Y \in \boldsymbol{Y}}}] = \sum_{U \in \boldsymbol{U}^i} \mathbb{E}_{(\hat{\pi}^i, \boldsymbol{\pi}^{-i})}[U_{\{Y_{\boldsymbol{\pi}|\boldsymbol{w}_{Y'}}\}_{Y' \in \boldsymbol{Y}'}}]$. so the inequality still holds for $\boldsymbol{Y}'$. So $\boldsymbol{Y}$ is not subset minimal and we have a contradiction. The proof that there must be a path from D^i to X is analogous: if there is no path from D^i to X, $X_{\boldsymbol{\pi}} = X_{(\hat{\pi}^i, \boldsymbol{\pi}^{-i})}$ for any π^i and $\hat{\pi}^i$. So no minimal $\boldsymbol{Y}$ contains X. □

Proof of completeness result for intention (theorem 6.3).

Proof. Suppose, without loss of generality (WLOG), that the path has the form $(D^i, X_1, ...X_n, U^i)$ with $X = X_k$ for some $k \in \{1, ..., n\}$ (additionally, any of the X_k may be a decision variable of another agent). Suppose (WLOG) that the domains of each variable in the path are ordered s.t. each variable has a max and min value ($|dom(V)| > 1$ by supposition). Consider the parameterisation $\boldsymbol{\theta}$ and $\boldsymbol{\pi}^{-i}$ s.t. $\Pr(X_{k+1} = max | X_k = max) = 1$ and similarly for "min". In addition, suppose $\Pr(X_1 = max | D^i = max) = 1$ and $\Pr(U^i = max(dom(U^i)) | X_n = max) = 1$ (similarly for "min"). If i has multiple utility variables, assume the others are constant under $\boldsymbol{\theta}$. Finally, take $\pi^i(D^i = max) = 1$ and $\hat{\pi}^i(D^i = min) = 1$. Now, under π^i, $U^i = max(dom(U^i))$ and under $\hat{\pi}^i$ $U^i = min(dom(U^i))$, and $U^i_{X_\pi}(\hat{\pi}) = max(dom(U^i))$. So i intentionally causes $X = max$ with π^i wrt $\hat{\pi}^i$ in every setting. □

Proof of soundness criteria for deception (theorem 3.8).

Proof. Deception is intentional by definition 3.7, so clearly the criteria for intent hold. In addition, there must be some X such that T does not observe X: by proposition 3.2 (an agent does not have a false belief about any proposition constituted only by variables which they observe). Since deception requires that T has a false belief about some proposition, there must be some variable constituting that proposition which is unobserved by T. □

Proof of completeness result for deception (theorem 3.9). We show this for the binary variable case – the general proof is analogous to the proof of theorem 6.3.

Proof. Suppose we have a graph with binary variables satisfying the criteria. Take the parametrisation such that $U^S = 1$ if $D^T = 1$ and 0 otherwise; $U^T = 1$ if $D^T = D^S$ or $D^T = X$ and 0 otherwise. Take the proposition $X = 1$, and the policy $\pi^T(\phi)$ for T such that $D^T_{\phi=\top} = 1$ and $D^T_{\phi=\bot} = 0$ (so D^T responds to observation of ϕ, satisfying the second condition for belief). Take the policy for T: π^T s.t. $D^T = D^S$. Now take π^S: $D^S = 1$ and $\hat{\pi}^S$: $D^S = 0$. Now, 1. S intentionally causes $D^T = 1$ with π^S w.r.t. $\hat{\pi}^S$ in every setting. 2. Additionally, $D^T = D^T_{\phi=\top}$, so T believes $X = 1$ in all settings, including when it is false. 3. Finally, suppose S does not respond to ϕ, and does not have any belief about it. So all the conditions for deception are met. □

6.1.5 PSO algorithm

Algorithm 1 presents the path-specific objectives (PSO) algorithm adapted from Farquhar et al. [31].

Algorithm 1 Path-Specific Objectives

Input: An SCG $\mathcal{M} = (\mathcal{G}, \boldsymbol{\theta})$, graphical criterion $\mathcal{C}$, policies $\boldsymbol{\pi}^{-i}$, natural distributions $\mathcal{N}$.
Output: PSO-optimal policy π^i

1: Reduce $\mathcal{G}$ to $\mathcal{G}'$ using $\mathcal{C}$.
2: Impute policies $\boldsymbol{\pi}^{-i}$ and natural distributions from $\mathcal{N}$ to those variables with fewer parents in $\mathcal{G}'$ to obtain $\boldsymbol{\theta}'$.
3: Train an agent in $\mathcal{M}' = (\mathcal{G}', \boldsymbol{\theta}')$ to obtain policy π^i.

6.1.6 Fixing problems with HKW intention

In this section we adapt H&KW's definitions of intention to SCGs. We demonstrate problems with their definitions and show that our definition of *intentionally cause* fixes these problems. In order to do this we adapt the notion of *actual causality* to SCGs.

Our definition of intentionally cause is given by definition 3.4.

HKW definitions

H&KW define the *intention to influence* a variable in structural causal models, and utilize this to define the *intention to bring about* a particular outcome of a variable. Our definition of intentionally cause is inspired by H&KW's intention to influence, and fixes important problems with their definition of intention to bring about. In this section, we adapt the H&KW definitions to SCGs, and generalise them from intention with a decision to intention with a policy. We present an example to show how our definition fixes problems with the H&KW definitions.

The intuition behind *intention to influence* is the same as for our intentionally cause. What it means for an agent to intentionally influences a variable X is that, if the effect the agent has on X was guaranteed to to happen anyway, then they would not mind choosing an alternative policy.

Definition 6.4 (Intention to influence)**.** Under $\boldsymbol{\pi} = (\pi^i, \pi^{-i})$, agent i *intends to influence* $\boldsymbol{X} \subseteq \boldsymbol{V}$ with policy π^i w.r.t. alternative policy $\hat{\pi}^i$ if there exists a subset-minimal $\boldsymbol{Y} \supseteq \boldsymbol{X}$ satisfying:

$$\sum_{U \in \boldsymbol{U}^i} \mathbb{E}_{\boldsymbol{\pi}}[U] \leq \sum_{U \in \boldsymbol{U}^i} \mathbb{E}_{(\hat{\pi}^i, \boldsymbol{\pi}^{-i})}[U_{\boldsymbol{Y}_{\boldsymbol{\pi}}}]. \tag{1}$$

H&KW build on intention to influence to define *intention to bring about* some particular outcomes. We first adapt intention to bring about to SCGs, and then present an example which shows that there are several problems with this definition, and explain how our definition of intentionally cause fixes